1 The viscosity of a fluid η is determined by the equation

$$\eta = \frac{\pi P r^4}{8l\left(\frac{V}{t}\right)}$$

where $\frac{V}{t}$ is the volume rate of flow of the fluid through a horizontal pipe,
l is the length of the horizontal pipe,
r is the internal radius of the pipe,
P is the pressure difference across the pipe.

(a) Show that when expressed in base SI unit, the units for η is kg m^{-1} s^{-1}.

[2]

(b) In an experiment to determine η for water, a student quoted his result as 1.137×10^{-3} kg m^{-1} s^{-1}.

He estimated the percentage uncertainties in his measurement of $\frac{V}{t}$, l, r, P as $\pm 3\%$, $\pm 2\%$, $\pm 0.5\%$ and $\pm 5\%$ respectively.

(i) Calculate the percentage uncertainty in the value of η.

percentage uncertainty = ... % [2]

Examiner's Use

(ii) Hence, express the value of η together with its absolute uncertainty. Give your answer to an appropriate number of significant figures.

η = (..... $\pm$ .....) $\times$ 10^{-3} kg m^{-1} s^{-1} [1]

[Total: 5]